

Graphical Abstract Text

Subject-Wise Evaluation Exposing Identity Leakage in Actigraphy-Based Mental-Health Prediction

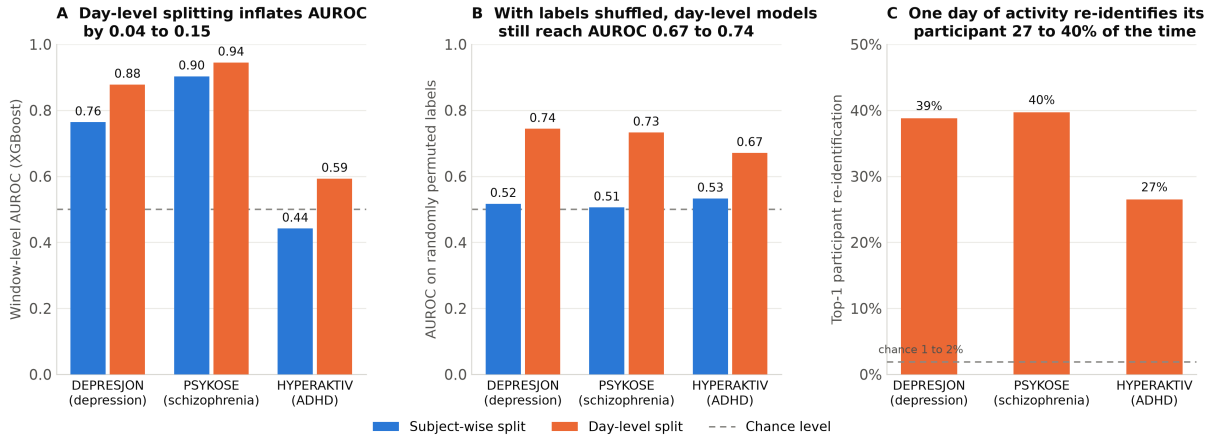
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Table-of-contents caption (50 words). Actigraphy models for depression, schizophrenia and ADHD lose 0.04 to 0.15 AUROC when evaluated subject-wise instead of day-wise. Day-wise models score above chance on shuffled labels and re-identify participants from one day at 27 to 40 percent, showing they learn identity; published accuracies above the honest band used day-level splits.

Accompanying image (uploaded separately as `graphical_abstract.png`, 3900×1980 px, 300 dpi):

Day-level evaluation of actigraphy models scores participant identity, not illness

Same features, same models, four public cohorts (DEPRESJON, PSYKOSE, HYPERAKTIV, OBF-Psychiatric); only the unit of cross-validation differs



Leakage-safe protocol: subject-wise splits, frozen cross-cohort transfer, calibration, permutation and re-identification checks. Of 28 published models on these cohorts, 21 split by day or did not say; every accuracy above the honest band coincides with day-level splitting.